

Credit Card Default Prediction on Kaggle Dataset

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Abstract

This study addresses the critical business problem of predicting credit card default for financial institutions. We applied and compared five machine learning models—Logistic Regression, Decision Tree, Random Forest, XGBoost, and a Cost-sensitive Ensemble—on the Kaggle "Default of Credit Card Clients" dataset. The results indicate that ensemble methods, specifically XGBoost and Random Forest, provide strong predictive performance with an Area Under the Curve (AUC) ranging from 0.76 to 0.78. Furthermore, a new approach called cost-sensitive ensemble was developed to reflect the high cost of a false negative (missing a defaulter). This ensemble dramatically increases sensitivity (up to 0.940). The model outputs are interpreted from a financial mathematics perspective as the Probability of Default (PD), a key input for Expected Loss calculation. For future work, we could perform more hyperparameter tuning or feature engineering, and extend this framework to other credit products and time periods.

Keywords: Credit Card Default; Probability of Default; Machine Learning; XGBoost; Random Forest; Cost-Sensitive Ensemble; Financial Risk Management.

1. Introduction

Credit risk management is fundamental to the stability and profitability of financial institutions that issue credit products, such as credit cards. The core operational challenge faced by these institutions is the accurate assessment of a client's solvency. Specifically, banks are highly motivated to predict whether a credit card client will default in the following month. This task is structured as a binary classification problem: a client defaulting is designated as $Y = 1$, and not defaulting as $Y = 0$.

From a financial mathematics perspective, the output of this predictive model serves a critical role. It is interpreted as the Probability of Default (PD), which is a key component in calculating the Expected Loss (EL) for a credit portfolio, defined by the formula: $EL = PD \times LGD \times EAD$, where LGD is the Loss Given Default and EAD is the Exposure at Default. Therefore, developing accurate and reliable PD models directly contributes to risk-based pricing, capital allocation, and regulatory compliance.

2. Literature Review

2.1 The Strategic Importance of Credit Default Prediction

Effective credit risk management is a cornerstone of financial stability. The central challenge involves accurately classifying applicants or existing customers into 'good' and 'bad' risk classes, a process formally known as credit scoring (Hand and Henley, 1997). With the dramatic growth of consumer credit, these statistical methods have become increasingly vital (Hand and Henley, 1997).

The output of a credit scoring model is the Probability of Default (PD), which holds a critical position in financial risk frameworks. The PD is a key input for calculating the Expected Loss (EL), defined by the relationship: $EL = PD \times LGD \times EAD$ (Thomas, 2009). Beyond merely minimizing defaults, modern lending objectives have expanded to include maximizing profits, pricing loans based on risk, and complying with regulations like the Basel Capital Accord (Thomas, 2009). Thus, accurate PD estimation is essential for strategic decision-making and portfolio management.

2.2 Mainstream Methodologies for PD Estimation

Since last century, a wide range of statistical methods have been applied to credit scoring.

2.2.1 Traditional Statistical Methods

Historically, Logistic Regression (LR) has served as the baseline model for PD prediction. Its popularity stems from its statistical robustness, ease of implementation, and its high interpretability—the relationship between input features and the log-odds of default is transparent and easy to explain (Hand and Henley, 1997).

2.2.2 Tree-Based Ensemble Learning

More recently, research has heavily shifted towards machine learning ensembles to improve predictive accuracy, particularly the ability to capture complex relationships within the data. Random Forest is a combination of tree predictors that leverages bootstrap sampling and random feature selection to enhance stability and reduce generalization error (Breiman, 2001). Random forest yields error rates comparable to advanced boosting methods while offering greater robustness to noise.

Friedman (2001) developed the concept of Gradient Boosting, framing function estimation as a numerical optimization problem using a steepest-descent "boosting" paradigm. This approach, particularly when utilizing regression trees (TreeBoost), yields robust and competitive procedures for classification.

As a scalable implementation of gradient boosting, XGBoost has become a dominant force in machine learning challenges involving tabular data (Chen, 2016). Its highly optimized architecture allows it to effectively handle large datasets (Chen, 2016). XGBoost is widely recognized in the field for its superior discriminative power, often yielding the highest Area Under the Curve (AUC) in classification tasks.

2.3 The Trade-off: Accuracy vs. Business Cost Sensitivity

While metrics like AUC provide a measure of the model's overall discriminative ability (ranking clients by risk), they often fail to capture the real-world asymmetric misclassification costs inherent in credit risk. In banking, the cost associated with a False Negative (FN) (accepting a client who defaults) is obviously higher than the cost of a False Positive (FP) (rejecting a client who would not have defaulted).

Standard models often use a default classification threshold (e.g., $p = 0.5$) that optimizes for overall accuracy, which may result in a sub-optimal outcome from a business risk perspective. To address this, the decision-making process must incorporate cost-sensitive learning by adjusting the probability threshold (Thomas, 2009).

This study directly addresses this trade-off by comparing the pure discriminative power of highly accurate models (Random Forest and XGBoost) against a Cost-sensitive Ensemble. The goal of the latter is not to maximize balanced accuracy but to drastically increase Sensitivity—the model's ability to correctly identify defaulters—by deliberately lowering the classification threshold. In institutions with highly conservative risk management strategies, minimization of expensive false negatives is prioritized over overall classification accuracy.

3. Dataset Description and Preprocessing

3.1 Dataset Description

The dataset used for this study is the UCI Default of Credit Card Clients dataset from Kaggle. The objective is to predict the binary target variable, `default_next_month`, which indicates whether a client will default on their credit card payment in the following month.

The data was initially partitioned into a training set (21,001 rows) and a test set (8,999 rows). The default rate in the designated test set was approximately 22%. The predictor variables fall into three main categories: Demographics (SEX, AGE, EDUCATION, MARRIAGE), Credit and Bill Information (LIMIT_BAL, monthly bill amounts (BILL_AMT1 through BILL_AMT6), and monthly payment amounts (PAY_AMT1 to PAY_AMT6)), and Repayment Status indicators for the past six months (PAY_0 to PAY_6).

3.2 Data Preprocessing and Cleaning

Several steps were executed to clean and prepare the raw data for modeling: The first step is to name the target. The identifier column ID was dropped. The original target variable was renamed to `default_next_month`. The second step is to address type conversion. The target was recoded as a factor, with 0 representing no default and 1 default. Categorical variables (SEX, EDUCATION, MARRIAGE, and PAY_0 through PAY_6) were all converted to factors. The third step is to do categorical value cleaning. To standardize and consolidate small groups, several categorical variable codes were grouped. For EDUCATION, the rare codes 0, 4, 5, and 6 were grouped as "others." The fourth step is to analyze missing data. Fortunately, this dataset contained no missing values. The last step is to training and test split. The cleaned data was partitioned into a 70% training set and a 30% testing set using stratified sampling. This ensured that the proportion of default cases was maintained consistently across partitions, with the test set reserved for final performance evaluation.

3.3 Exploratory Data Analysis

The analysis of the distribution of the credit limit, `LIMIT_BAL`, indicates a highly right-skewed profile: most clients have low or medium credit limits, while only a few have very high credit limits. In addition, exploratory data analysis highlights a significant correlation between high delinquency codes and future default risk: clients whose records show a payment delay of two months or more (i.e., codes ≥ 2) demonstrate an increased probability of defaulting in the subsequent month. Consequently, these historical repayment variables should be given importance in model construction as expected. Last but not least, strong correlation between bill amounts and payment amounts matters when interpreting logistic regression later.

3.4 Specific Preparation for XGBoost

Given that the XGBoost framework necessitates a numerical matrix input, additional data preparation was performed for this specific model. Rows with NA values were removed using `na.omit`, design matrices were built using `model.matrix`, which performs one-hot encoding for factor variables. Then, factor levels were ensured to maintain consistent between the training and test data. This setup allows us to compare all models fairly on a common test set.

4. Methodology

4.1 Model Selection and Implementation

The prediction of the Probability of Default (PD) was achieved through the comparative application of five distinct classification models. The suite included a foundational statistical model, two established tree-based ensemble methods, a state-of-the-art boosting framework, and a customized cost-sensitive approach designed with novelty. Model 1 is logistic regression, which is a baseline probability-of-default model. Model 2 is a decision tree built using the `rpart` package. Model 3 is a random forest, which combines multiple trees to reduce variance. Model 4 is XGBoost, a gradient boosting method designed for tabular data. Model 5 is cost-sensitive ensemble designed by our team. It was developed by averaging the predicted probabilities from the Logistic Regression, Random Forest, and XGBoost models. This ensemble's unique purpose was to select a classification threshold that directly minimizes the expected cost of errors, aligning the model's output with asymmetric financial risk management objectives.

4.2 Model Evaluation Metrics

Model performance was rigorously evaluated using several metrics derived from the confusion matrix, ensuring a comprehensive assessment. Accuracy is the overall proportion of correct predictions. As a result of the inherent class imbalance in the dataset, performance on the minority class (recall for the default class) was monitored through Sensitivity, which quantifies the rate at which true defaulters are accurately identified. This metric is particularly important for the cost-sensitive approach. Conversely, Specificity measures the recall for the non-default class. Balanced accuracy is the average of sensitivity and specificity, which provides a reliable measure unbiased by class distribution. The primary measure of the model's discriminative power was the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) curve. A higher AUC indicates a greater ability to correctly rank riskier clients above safer clients.

5. Model Results and Performance Evaluation

The predictive performance of the five developed models was evaluated on the independent test set. The models were primarily assessed on their discriminative power (AUC) and their ability to accurately classify both defaulters (Sensitivity) and non-defaulters (Specificity), using Balanced Accuracy to summarize overall performance in the face of class imbalance.

5.1 Model 1: Logistic Regression

The Logistic Regression (LR) model, as the baseline, demonstrated reasonable discriminative performance with an AUC of approximately 0.7631. Its Balanced Accuracy was recorded at 0.6552. The performance of the LR model established a good benchmark for the study.

5.2 Model 2: Decision Tree

The single Decision Tree model yielded an Accuracy of approximately 0.8202. However, its discriminative power was significantly lower than the baseline model, with an AUC of only 0.6456. This model achieved very high Specificity (around 0.959), but suffered from extremely low Sensitivity (around 0.333). Consequently, the Decision Tree missed a substantial proportion of actual defaulters, which represents a high-risk outcome for a financial institution.

5.3 Model 3: Random Forest

The Random Forest model, an ensemble approach, exhibited strong performance, outperforming the single Decision Tree. It achieved an Accuracy of 0.8195 and a strong AUC of 0.7604. This approach raised Sensitivity to approximately 0.385 while maintaining high Specificity (around 0.943). Its Balanced Accuracy was the highest among the base models at 0.6640.

5.4 Model 4: XGBoost

The XGBoost model, a gradient boosting framework, demonstrated superior discriminative capability. It achieved an Accuracy of approximately 0.8195 and the highest AUC among all single models at 0.7777. The model also maintained strong performance, with a Sensitivity of approximately 0.374, a Specificity of 0.946 and a Balanced Accuracy of 0.6598.

5.5 Model 5: Cost-Sensitive Ensemble

The Cost-Sensitive Ensemble was designed not to maximize overall accuracy, but to minimize the cost of errors by prioritizing the capture of defaulters. The ensemble probability was defined as the average of the predicted probabilities of logistic regression, random forest, and XGBoost models:

$$\hat{p}_{ens} = \frac{\hat{p}_{logit} + \hat{p}_{rf} + \hat{p}_{xgb}}{3}$$

The classification threshold t ($t = 0.10$ was optimal.) was used to minimize the cost function $Cost = C_{FN} \cdot FN + C_{FP} \cdot FP$, where the cost of a False Negative (C_{FN}) is much greater than the cost of a False Positive (C_{FP}). This model achieved an exceptionally high Sensitivity of approximately 0.940, at the cost of overall classification metrics: Accuracy dropped to 0.4283 and Specificity fell to 0.283. The Balanced Accuracy was 0.6114, while the AUC remained robust at 0.7780.

5.6 Model Comparison Summary

Model	Accuracy	Balanced Accuracy	Area Under the Curve (AUC)
Logistic Regression	–	0.6552	0.7631
Decision Tree	0.8202	0.6456	0.6456
Random Forest	0.8195	0.6640	0.7604
XGBoost	0.8195	0.6598	0.7777
Cost-sensitive Ensemble	0.4283	0.6114	0.7780

Table 1: Model Performance on the Test Set Under different Metrics

As shown in Table 1, both the Random Forest and XGBoost models are robust, yielding Balanced Accuracy scores around 0.66 and high AUC values around 0.77. The results highlight a trade-off between statistical accuracy and missing defaulters: while the Decision Tree offers simple explanation, its low AUC and poor Sensitivity render it less reliable for risk assessment. In contrast, the Cost-sensitive Ensemble achieves high Sensitivity, but sacrificing accuracy and specificity.

6. Discussion and conclusion

The practicability of the models lies in the interpretation of their output as the Probability of Default (PD), integral to the calculation of Expected Loss (EL), allowing financial institutions to execute risk-pricing and set dynamic PD thresholds under risk appetite or regulatory mandate. Furthermore, the cost-sensitive ensemble performs well in high-exposure scenarios where minimizing the cost of False Negatives is paramount.

In conclusion, a full data analysis and modeling pipeline was successfully completed on the credit card default dataset. The comparative analysis of the five classification models shows that XGBoost and Random Forest delivered the most robust predictive performance, achieving an AUC between 0.76 and 0.78. Sensitivity rose dramatically with cost-sensitive ensemble model, rendering it an effective tool for high risk-aversion. The limitation is that this study was conducted on a single static dataset, future work could test robustness across different macroeconomic conditions. Future work could also focus on calibrating the predicted PDs for real-world reliability, conducting hyperparameter tuning and feature engineering.

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